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Experience-based group education in Type 2 diabetes A randomised controlled trial

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Abstract

Few studies have demonstrated an effect of educational interventions on glycaemic control in persons with Type 2 diabetes longer than 3–6 months after baseline. We aimed to investigate the effectiveness of an experience-based group educational programme 24 months after baseline and to pinpoint mediators that might play a role in achieving desired metabolic outcomes. We conducted a randomised controlled trial inviting self-referred persons with Type 2 diabetes ($N = 77$ randomised). The pharmacist-led, year-long intervention was based on participants' experiences of glucose regulation during the monthly group discussions. We measured HbA_{1c} at 0, 6, 12, and 24 months and a questionnaire was administered at baseline and final follow-up. Our findings indicated that participating in the intervention programme significantly decreased HbA_{1c} by 0.4% at 24 months after baseline. Initial HbA_{1c}, satisfaction with own diabetes-related knowledge, and treatment were found directly related to glycaemic outcomes. The intervention group exercised more in order to lower blood-glucose levels and was also more able to predict current blood-glucose levels before measuring it. Experience-based group education was effective in decreasing participants' HbA_{1c} 1-year after completed intervention. Early effect of the intervention was followed by relapse after 12 months and a new, significant decrease at 24 months; this dual course implies that follow-up of educational interventions should involve several consecutive measurements to capture possible late effects. Both biomedical and subjective factors played a role in accounting for the variance of HbA_{1c} at 2-year follow-up after baseline.

Keywords: Patient education; Diabetes mellitus; Randomised controlled trial; HbA1c

1. Introduction

Patient education is a well-established component of modern diabetes management [1]. However, there is no consensus on how effective educational efforts should be planned and put into practice to achieve long-term effects.

In a recent meta-analysis on randomised trials on patient education in Type 2 diabetes, the authors demonstrated that the longer the follow-up after an educational intervention, the lesser the effect on HbA_{1c} [2]. The greatest effects were observed between 0 and 3 months after the intervention, to fade away entirely by 12 months. Few studies demonstrate an effect on glycaemic control longer than 6 months after educational intervention, whereas most educational programs achieve these early effects [3]. Consequently, it would appear that most interventions will help in the short run, but as time passes, whatever caused the initial effect seems to vanish in the reality of everyday life [2,4]. The question that arises then is whether it is possible to achieve long-term glycaemic control through patient education by choosing better intervention targets.

R*eflection* and *understanding* are concepts that underline the pivotal role that persons with diabetes have in achieving effective daily management of their disorder. Acknowledging the individual's competence to make decisions about everyday care yields a sense of *autonomy*, which is assumed to help individuals "take charge of their own diabetes" [5]. This initiative, in turn, results in the individual acting as an equal partner in the planning and delivery of diabetes care. In this perspective, the concept of *empowerment* can be viewed as both the goal and the means of patient education in diabetes. Supporting these assumptions is the evidence that *internal locus of control* [6], reflecting the individual's sense of power to influence his or her situation, and health-care staff attitudes conducive to autonomy [7] have predicted a decrease of HbA_{1c} in diabetes educational programs. In a randomised controlled study of persons with arthritis, *self-efficacy* was distinguished as the mediating

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factor in educational intervention and improved disease outcomes along with decreased physician costs [8].

Another intervention target with as yet unknown potential is the role of body awareness that has been studied in relation to Type 1 diabetes where training programs for patient recognition of hyper- and hypoglycaemia have proved useful for daily disease management [9]. Price described the process [10] in which persons with Type 1 diabetes learned to manage their illness during the first year after diagnosis by developing skills of "body listening" and interpreting the signs and clues their bodies gave them. Whether training body awareness has the potential of improving glycaemic control has not been examined, however. In an explorative interview study the interviewees described a "feeling in the body," a way of knowing what was going on with their blood-glucose levels and why, as a helpful instrument for controlling their diabetes [11].

In this paper we present a randomised controlled trial of an experience-based group educational program. The intervention targeted several of the above mentioned subjective aspects of diabetes self-management and our aim was to investigate the effectiveness of the intervention as well as attempt to pinpoint mediators [12] that might have played a role in achieving desired metabolic outcomes. We hypothesised that body awareness and a capability to make a correct guess of the current blood-glucose level before measuring it would predict better glycaemic control. Other hypotheses tested included positive effects of exercise and decreased body weight and negative effects of feelings of loneliness, high body mass index (BMI) and feelings of anxiousness for diabetes-related complications [13]. Educational background, age, and sex were basic variables which we planned to initially include in all statistical testing. Planned analyses included one-way ANOVA, regression- and repeated-measures analyses, and independent sample *t*-tests.

2. Methods

2.1. The intervention

The intervention was a 12-month long group educational program led by specially trained pharmacists, assisted by a diabetes nurse specialist on the first two occasions. The programme had been pilot-tested [14] and we also conducted a trial for implementing the intervention at a number of pharmacies around Sweden to test its feasibility for mass education [13].

The training of the pharmacists (by co-author U.R.) to become facilitators comprised a 3-day intensive course where the main objective was to convey the pedagogical principle of the program, experience-based learning, by applying it to the participating pharmacists. Throughout the course the pharmacists monitored their blood-glucose levels, did the shopping for lunch and snacks, prepared meals, and went on walks after meals to test the effects of exercise on blood-glucose levels. The educational materials used were identical to what the pharmacists were to use with program participants: a video on how to "live well" with diabetes, exemplifying lifestyle changes made by those interviewed; a dice game where questions had to be answered, but where no set answers were available, but had to be negotiated by players; and a booklet or guide on "how to manage your diabetes", using the metaphor of a rowboat, which is first difficult and scary to control, but with time possible to master. The booklet also contained logs of imaginary people who had some typical faults in their diet or treatment and were used to stimulate discussion of more appropriate routines. The book further included information about diabetes complications (including both female and male sexual dysfunction) and a personal plan for follow-up visits. The pharmacists were instructed not to intervene with participants' medical regimens, but refer them to their medical team when glucose control seemed unsatisfactory despite adequate diet and exercise. Continuous back-up and support was provided to the pharmacists with regular follow-up group meetings every 6 months. The pharmacists also kept a diary for each participant to record their learning experience throughout the program.

The goal of the educational program was to reinforce the participants' experiences and use these experiences as a basis for the acquisition of practical skills needed for self-management of diabetes [15]. Participants were encouraged to "experiment" with different nutritional components and exercise and monitor their blood-glucose reactions as a means to promote experience-based learning. The groups met once a month during a year and the self-monitoring diaries of participants were shared with the group and comprised an important foundation for discussions. The guiding pedagogical principle throughout the group sessions was that any questions raised should be solved by the group rather than by the group leader. During the intervention, the practical aspects of diabetes management, such as choice and preparation of food, performing self-monitoring tasks, and walks or jogs to decrease blood-glucose levels, were handled during the group sessions. The educational program was also geared to provide the participants with support for dealing with the emotional aspects of the disease.

Length of intervention was 1-year and planned follow-up was 1-year after completed intervention, i.e. 2 years after baseline. The control group was assigned to a waiting list of 2 years. Then they were invited to participate in the educational program.

2.2. Outcome measures

The principal outcome measure was HbA_{1c}, measured at baseline and after 6, 12 and 24 months in both the intervention and control groups. Power calculations resulted in 18 participants per group necessary to detect a decrease of 1% unit in HbA_{1c}, (e.g., 7.2– 6.2%) with $\alpha = 0.05$ and $\beta = 0.1$ using two-tailed testing [16]. Because a higher number of

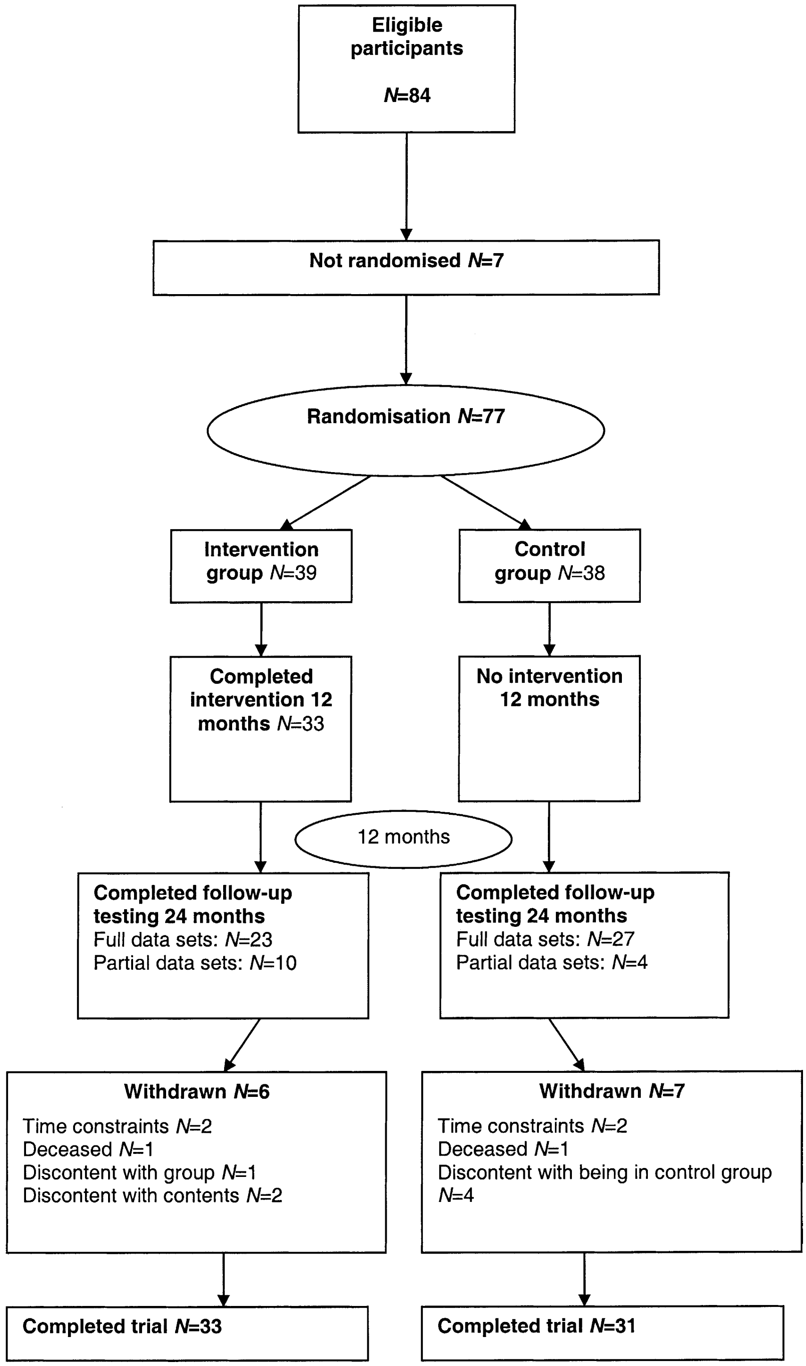

Fig. 1. Flow-chart of events in the study. From recruiting participants through randomisation and study completion.

participants were required for other explanatory variables we wanted to test and because we counted on at least a 20% drop-out rate, we recruited 84 participants in total. The events of randomisation through follow-up and completion along with drop-outs and the reasons for this are depicted in the flow-chart (Fig. 1).

A questionnaire was administered at the start of the study and 2 years after baseline (Table 1). Participants were invited to comment on all of the items related to personal perceptions about the disease; quotes from these comments are used to further explain and/or exemplify certain results. We have also performed a full-scale qualitative analysis of these texts, the results of which will comprise a separate article.

2.3. Participants

Participants were self-referred, responding to advertisements in local newspapers and flyers distributed at the GP's office and the Stockholm Diabetes Association. The Ethics Committee of Uppsala University Medical Faculty gave its

Table 1
Items in the two questionnaires

Questionnaire at T_0 Descriptive variables	Common items Descriptive variables	Questionnaire at T_{24} Descriptive variables
Age and sex (II)	Weight and height (II)	Has your weight changed during the past 2 years?[a]
Year of diabetes diagnosis (II)	Treatment (II)	How many times did you visit a health practitioner during the past 12 months?[b]
Other diseases than diabetes	Marital status (III)	Has the treatment of your diabetes changed during the past 2 years?[a]
Employment Type (III)	Employment status (III)	Have you been admitted to hospital consequent to your diabetes during the past 2 years?[a]
	Complications	Have you lost a close friend or relative during the past 2 years?[a]
		Have you changed your exercising habits during the past year?[a]
		Do you smoke (Too few smokers in the group to be included in analyses)[c]
	Subjective items	Subjective items
	Do you feel lonely? (III)	Do you exercise regularly with the aim to affect your blood sugar?[a]
		Are you satisfied with your knowledge about diabetes?[a]
		Are you worried about complications you can develop because of diabetes?[a]
		I can predict my blood-glucose value rather securely before measuring it[d]
		I can “feel in my body” where my current blood sugar level is.[d]
		I feel in control of my blood sugar and can keep it on the right level.[d]
		My blood sugar changes irrespective of what I do to influence it.[d]

Common items indicate repeated measurements at t_0 and t_{24}. The number of the table where data are presented in the article is in parentheses where appropriate. Question types are indicated in the footnote.

[a] Yes; no; (I don’t know). If yes, please detail!

[b] Open-ended question.

[c] Closed question with given options.

[d] Scale 1–7.

approval to the study. Two inclusion criteria for registering persons applying for participation were used: participants had to be diagnosed with Type 2 diabetes and, if treated with insulin, only for 2 years or less. Although the educational program was originally designed for persons who recently received their diabetes diagnosis, evidence from an earlier pilot study indicated that even persons with diabetes of longer duration could benefit from the intervention [14]. The exclusion of persons with long-term insulin treatment was determined based on reports from the study circle leaders who felt that dietary and exercise interventions did not lead to immediately demonstrable effects for this group of participants. This could be because, until recently, those receiving insulin therapy in Type 2 diabetes were persons whose diabetes was insufficiently controlled under a long period before initiation of the insulin treatment.

In order to participate in the randomisation, participants had to leave an initial HbA_{1c} measurement, complete a questionnaire, and give their informed consent to participate in the study. If any of these items were missing and no completion was made despite reminders, the person was excluded from the randomisation process. This procedure led to the exclusion of seven persons (Fig. 1).

For those participants eligible for randomisation, the informed consent sheet and the questionnaire were put into an unmarked envelope, one for each participant. The identical envelopes were then put into a box. Each time 20 complete sets of participant items were collected, randomisation was performed. An assistant mixed the envelopes in the box, took them out one at a time, and randomly placed them into two piles. A third person, acting as a witness, pointed out which pile should be allocated to the intervention group and which pile to the control group. Each participant was then assigned a code, beginning with a different letter for the intervention (I) and control (C) groups. A research assistant then sent a standard letter to each participant, indicating which group the participant was assigned and communicating the result of the HbA_{1c} measurement.

3. Results

The intervention and control groups did not differ on age, sex, BMI, employment, and marital status (Tables 2 and 3). The intervention group, however, was found to have longer diabetes duration compared with the control group (5.9 years versus 2.6 years). Probably therefore, the intervention group also included more persons who were on both oral hypoglycaemic agents and insulin.

3.1. Long-term glycaemic control (HbA_{1c})

The following differences were calculated and used during analyses: that between HbA_{1c} t_0 and t_6 ($\mathrm{Diff}_{t0\text{–}6}$), HbA_{1c} t_0 and t_{12} ($\mathrm{Diff}_{t0\text{–}12}$), and HbA_{1c} t_0 and t_{24} ($\mathrm{Diff}_{t0\text{–}24}$). One-way ANOVA for the three differences [($\mathrm{Diff}_{t0\text{–}6}$), ($\mathrm{Diff}_{t0\text{–}12}$), and ($\mathrm{Diff}_{t0\text{–}24}$)] showed that the intervention group decreased its HbA_{1c} values significantly more than the control group in the short-term follow-up between 0 and 6 months after baseline ($P = 0.05$). The control group did not show any change in its HbA_{1c} values between the 0 and 6-month period. Likewise, the long-term follow-up at 24 months after baseline demonstrated a prevailing decrease in HbA_{1c} in the intervention group ($P = 0.023$) compared

Table 2
Group statistics

Group	*N*	Mean	Standard deviation	Standard error	*t*-test for equality of means	d.f.	Significance 95% test (two-tailed)	Confidence interval of the difference
Age								
Control	31	66.5	10.7	1.9				
Intervention	33	66.4	7.9	1.4	0.037	62	0.970–4.6	4.8
BMI t_0								
Control	31	28.6	5.8	1.0				
Intervention	33	27.2	3.6	0.6	1.184	62	0.241–0.1	3.8
Duration								
Control	27	2.6	2.2	0.4				
Intervention	33	5.9	5.8	1.0	−2.774	58	0.007–5.6	−.9

Comparison of continuous variables: Age, initial body mass index (BMI), and duration of diabetes counted in years since self-reported diagnosis.

Table 3
Summary of four regression analyses based on original and adjusted data

Regression model	HbA_{1c} t_0	Intervention/control			
Outcome	For all models (*P*-value)	Non-adjusted original data (*P*-value)	Adjusted (*P*-value)	Non-adjusted, log-transformed data (*P*-value)	Adjusted log-transformed data (*P*-value)
HbA_{1c} t_6	<0.001	0.047	0.062	0.053	0.069
HbA_{1c} t_{12}	<0.001	0.240	0.256	0.261	0.303
HbA_{1c} t_{24}	<0.001	0.008	0.063	0.006	0.043

Initial HbA_{1c} and the dichotomous intervention/control variable were found to significantly affect outcomes at 6 and 24 months after baseline. This effect, however, varied for the intervention/control variable when the data was adjusted for duration and treatment as well as when logarithmically transformed data was used. Participating in the intervention program showed no effects for the measurements at ∼2 months in any of the models.

with the control group in which a non-significant increase was observed (Fig. 2). There was no disparity between the groups for the $t_{0–12}$ difference, i.e. the intermediate 12-month follow-up.

3.2. Regression and repeated-measures analyses

Because of the differences in duration and treatment between the intervention and control groups, models weighted

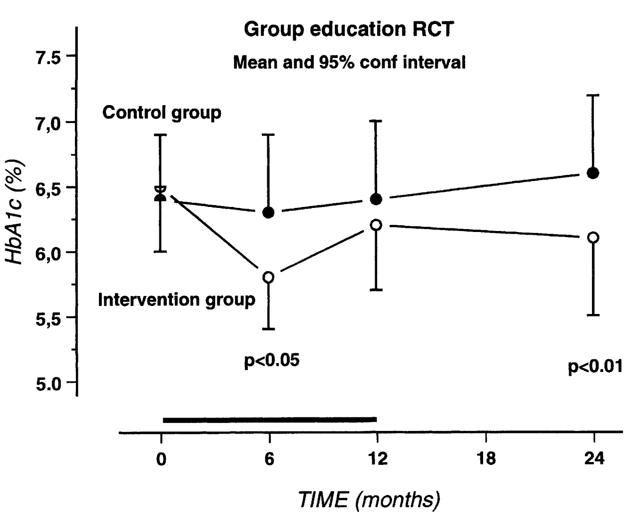

Fig. 2. Measures of HbA_{1c} in the intervention and control groups at baseline, after 6, 12, and 24 months. Means and 95% confidence intervals provided. Bold line on time axis (X) shows duration of intervention.

for these disparities were also tested. In the regression procedure, initial HbA_{1c} and the dichotomous intervention-control variable were found to significantly affect outcomes at 6 and 24 months after baseline before adjustment, but remained significant only for the 24-month outcome after adjustment was made for duration and treatment (Table 3). In a model adjusted for baseline HbA_{1c} only, being satisfied with own diabetes specific knowledge significantly influenced outcomes at the two year measure ($P = 0.03$), rendering the intervention variable insignificant. This factor remained significant in the fully adjusted model if the intervention variable was lifted out from the model, indicating an interaction where satisfaction with own diabetes-related knowledge could better predict outcomes than participating in the intervention per se.

The intervention produced no effects for the measurements at 12 months. Duration and treatment did not affect any of the outcome measures in either the weighted or the unadjusted models.

The repeated-measures analyses were conducted using SAS software. When checking the variance in the HbA_{1c} measurements, we observed higher residuals as a function of increasing HbA_{1c} values, i.e. the higher the HbA_{1c} values the greater the residuals. Therefore, ranked and logarithmic transformations of HbA_{1c} values were calculated and analyses performed on both the original and transformed values. Unstructured and symmetric covariance structures in the model were tested. However, unlike the regression analysis, the conclusions drawn here were very similar in the adjusted and unadjusted models. Thus, only the results based on the original datasets will be presented, if not otherwise indicated.

The following factors had a significant influence on the course outlined by repeated-measures of HbA_{1c}: initial HbA_{1c} ($P < 0.0001$), participating in the intervention group ($P = 0.010$), and treatment ($P = 0.0005$). Also, the interaction term satisfied with knowledge from participating in intervention was highly significant ($P = 0.006$), displacing the intervention variable to a non-significant position when included in the model. Duration was not significant in predicting outcomes. Effect of the intervention as a function of time was examined in a separate model, comparing least squares means, with significant effect only at 24 months after baseline ($P = 0.009$).

3.3. Univariate models

With the difference between HbA_{1c} t_0 and t_{24} ($\mathrm{Diff}_{t0\text{–}24}$) as dependent variable, participating in the intervention program ($P = 0.023$) and the initial HbA_{1c} value ($P = 0.001$) were factors that significantly influenced outcomes in the univariate correlational analyses. Age, sex, treatment, BMI, marital status, educational background, current employment status, other diseases and feelings of loneliness had no significant effect on outcomes in the univariate analyses. Other potential confounding factors influencing glycaemic control examined in this study with no significant effects were: changes in medical treatment of diabetes, hospital stay because of diabetes, the number of visits to health professionals, self-reported changes in exercising habits, and death of a close friend or relative.

We continued the investigation by establishing the possible causes of the different courses for the two groups. For each variable that altered the significant effect of the control-intervention group factor on delta HbA_{1c} ($\mathrm{Diff}_{t0\text{–}24}$) in the univariate model, we conducted a separate univariate analysis with this specific variable in the dependent position. The following factors (listed in the third column of Table 1) evidenced a significant difference between the control and intervention groups: being more satisfied with one's own knowledge about diabetes ($P = 0.008$), exercising more in order to affect blood-glucose levels ($P = 0.015$), and being able to predict current blood-glucose levels before measuring it ($P = 0.048$). Initial HbA_{1c} remained significant for the "exercise" and "being able to predict" variables in the dependent position. Factors without a detectable influence were an ability to "feel in my body" where the current blood-glucose level is, feeling in control of blood-glucose levels, and a feeling that blood-glucose levels change irrespective of personal efforts to influence it.

4. Discussion and conclusions

Our results indicate several points of interest. The most important implication of this study is that the experience-based educational intervention under study has produced a significant decrease in the HbA_{1c} levels of the intervention group. The intervention group had a greater decrease in HbA_{1c} both at 6 and 24 months after baseline, with the latter finding being significant in that few studies of educational programs have demonstrated long-term effects on metabolic control [2]. Those that have, on the other hand, reported a similar or lesser magnitude of effects as this study, where a 0.4% decrease in HbA_{1c} for the intervention group was observed 2 years after baseline. In a recent Italian study the control group deteriorated in HbA_{1c} whereas the intervention group maintained initial levels of glycosylated haemoglobin 4 years after baseline [17]. Yet another study from Denmark demonstrated a decrease in HbA_{1c} for the intervention group by 0.5% points after 8 years [18]. As Type 2 diabetes is still considered a chronic and progressive disease [19], a halt in disease progression, as measured by HbA_{1c} is to be considered a success; being able to revert the trend is certainly a very encouraging result, however it is achieved.

Another important implication of this study is the pattern of HbA_{1c} levels during the follow-up period. The effects of this experience-based educational intervention can be described by a dual course of events. An initial significant effect at 6 months after baseline was followed by a "rebound" at the 12-month follow-up, and yet another decrease in HbA_{1c} levels at 24 months after baseline. Many evaluations of educational or behavioural interventions cease their follow-up at 12 months, if not by 3–6 months after the intervention. Our results indicate that intervention effects may be postponed or have fluctuating dynamics, implying that follow-up should not be discontinued a year after baseline, but rather involve several measurements later on to capture a possible late effect of the intervention.

Because one of our aims was to attempt to pinpoint what might have been effective within the intervention, a third finding that deserves attention is the fact that both biomedical and subjective factors played a role in accounting for the variance in the differential between baseline HbA_{1c} and that measured at 2-year follow-up ($\mathrm{Diff}_{t0\text{–}24}$). Initial HbA_{1c} influenced outcomes in all our general linear models. This is in accordance with our earlier studies [13] as well as clinical experience. Treatment had a significant effect in the repeated-measures analyses, which most probably is a reflection of the influence of treatment modifications during the follow-up period. This is supported by the fact that no difference in treatment between the groups was seen at 24 months, whereas more participants had been on both oral hypoglycaemic agents and insulin in the intervention group initially. Participant's weight did not change significantly in any direction during the study period and BMI had no effect on outcomes as neither did feelings of loneliness or being anxious for diabetes-related complications.

With special interest directed to subjective variables, we found that being satisfied with own diabetes-related knowledge directly affected outcomes and in fact seemed to be a better predictor of decreased HbA_{1c} at 24 moths after baseline than participating in the intervention per se. Other variables found to significantly differ between the intervention and control groups were: exercising with a specific aim to lower blood-glucose and being able to accurately predict values before performing a glucose test.

4.1. Reliability and limitations

The participants in this study were self-referred, responding to ads in newspapers and flyers. This selection procedure introduced a systematic bias in the sample, presumably resulting in persons motivated to improve diabetes self-management. This observation also provides a probable explanation for the rather low mean initial HbA_{1c} of participants, with 52% under the WHO target value of 6.5% [20], which can be compared with a proportion of ~40% in a national sample of 10,000 persons [21]. On the other hand, randomisation occurred after the recruitment of participants so the bias, if any, was equally present for both intervention and control groups.

4.2. Practice implications

As we see it, the most important finding in our study is the significant role of subjective measures, reflecting personal perceptions of participants, in determining outcomes. Being more satisfied with one's own knowledge about diabetes and exercising to a greater extent in order to affect blood-glucose levels are not the same as increased knowledge about diabetes in general and exercising more per week. This distinction is clear from the participants' comments in the questionnaire: "*I know what kind of carbohydrates I can eat with least 'harm'...*" and "*I take a walk when I know I've eaten something 'wrong'*." We also believe that the ability to predict glucose levels accurately, which was significantly more common in the intervention group, is a product of an integrated knowledge comprised of information about dietary facts, earlier experience of adverse events, and the skill of listening to one's body. This latter finding underlines the role of body awareness in Type 2 diabetes, a phenomenon that prompts further study and a need to better conceptualise this potentially useful educational tool.

Other studies also indicate that, apart from predetermining demographic variables, highly subjective factors, such as perceptions, beliefs and attitudes will influence a patient's adaptation of behaviour, along with features of their social environment [22]. The importance of subjective and vaguely defined variables that relate to a personal understanding of and actions related to diabetes may be frustrating on the one hand, as it makes planning of standard educational interventions more difficult. On the other hand, it emphasises the great potential of experience-based pedagogy in educating persons with Type 2 diabetes and may provide part of the explanation as to why improving knowledge about diabetes per se is not enough [23] and why it is so difficult to conceptually capture and measure predictors of success in diabetes patient education [24]. Therefore, we recommend that evaluation of patient education programmes in the future should routinely include adequate assessment of subjective factors that potentially influence outcomes.

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References

[1] National Diabetes Advisory Board. Quality recognition for diabetes patient education programs: review criteria for national standards from the American Diabetes Association. Diab. Care 1986; 9:36–40.
[2] Norris S, Engelgau M, Venkat Narayan K. Effectiveness of self-management training in Type 2 diabetes. Diab Care 2001;24:561–87.
[3] Norris S, Lau J, Smith S, Schmid C, Engelgau M. Self-management education for adults with Type 2 diabetes: a meta-analysis of the effect on glycemic control. Diab Care 2002;25:1159–71.
[4] Brown S. Meta-analysis of diabetes patient education research: variations in intervention effects across studies. Res Nurs Health 1992;15:409–19.
[5] Anderson RM, Funnell MM, Barr PA, Dedrick RF, Davis WK. Learning to empower patients. Diab Care 1991;14:584–90.
[6] White N, Carnahan J, Nugent C, Iwaoka T, Dodson M. Management of obese patients with diabetes mellitus: comparison of advice education with group management. Diab Care 1986;9:490–6.
[7] Williams G, Freedman Z, Deci E. Supporting autonomy to motivate patients with diabetes for glucose control. Diab Care 1998;21:1644–51.
[8] Lorig K, Mazonson P, Holman H. Evidence suggesting that health education for self-management in patients with chronic arthritis has sustained health benefits while reducing health-care costs. Arthritis Rheum 1993;36:439–46.
[9] Cox D. Blood-glucose awareness training (BGAT-2). Diab Care 2001;24:637–42.
[10] Price M. An experimental model for learning diabetes self-management. Qual Health Res 1993;3:29–54.
[11] Jonsson A. ''Lära sig att bli den människa man måste bli" (''To learn to be the person one has to be"). C-uppsats. Stockholm: Stockhloms University, Department of Education 1995.
[12] Peyrot M. Behaviour change in diabetes education. Diab Educator 1999;25(S):62–73.
[13] Sarkadi A, Rosenqvist U. Field test of a group education programme for Type 2 diabetes: measures and predictors of success individual and group levels. Pat Educ Couns 2000;44:129–39.
[14] Sarkadi A, Rosenqvist U. Study circles at the pharmacy: a new model for diabetes education in groups. Pat Educ Couns 1999;37:89–96.
[15] Rosenqvist U, Theman J, Assal J-P. New developments in diabetes education: a call for a new task for the patient. Geneva: European Association for the Study of Diabetes (EASD) 1994.
[16] Lemeshow S, Hosmer D, Klar J, Lwanga S. Adequacy of Sample Size in Health Studies. New York: vol Wile, 1990.
[17] Trento M, Passera P, Bajardi M, Tomalino M, Grassi G, Borgo E, et al. Lifestyle intervention by group care prevents deterioration

of Type II diabetes: a 4-year randomised controlled clinical trial. Diabetologia 2002;45:1231–9.

[18] Gaede P, Vedel P, Larsen N, Jensen G, Parving H, Pedersen O. Multifactorial intervention and cardiovascular disease in patients with Type 2 diabetes. N Eng J Med 2003;348:383–92.

[19] United Kingdom Prospective Diabetes Study Group. Intensive bloodglucose control with sulphonylureas or insulin compared with conventional treatment and risk of complications in patients with Type 2 diabetes (UKPDS 33). Lancet 1998; 352: 837–53.

[20] Swedish National Board of Health and Welfare. Clinical guidelines for care and treatment of patients with diabetes mellitus. Sockholm 1996. Report No.: 4:2, December 1996.

[21] Swedish Department of Health and Welfare. National Diabetes Register. http://www.ldc.lu.se/NDR/arkiv/vedbo/hba1c_p2.htm 1999.

[22] DeWeerdt I, Visser A, Kok G, Van Der Veen E. Determinants of active self-care behaviour of insulin treated patients with diabetes: implications of diabetes education. Soc Sci Med 1990;30: 605–15.

[23] Bloomgarden Z, Karmally W, Metzger M, Brothers M, Nechemias C, Bookman J, et al. Randomized, controlled trial of diabetic patient education: Improved knowledge without improved metabolic status. Diab Care 1987;10:263–72.

[24] Day J. Diabetic patient education: determinants of success. Diab Met Res Rev 2000;16:S70–4.